ClockIn 🔒

Track what you *actually* did, not just what you planned. Fast session tracking, history, editing, and lightweight reflection.

The Problem

Lost Time

Students & professionals plan their day, but rarely know where the time actually went.

Friction Kills Habits

Tracking must be fast and interruption friendly to stick.

Mobile-First

Quick start/stop interactions throughout the day demand a phone-first experience.

What ClockIn Does

Tag-based sessions with a dashboard showing today's totals at a glance.

01

Pick a Tag

Study, Class, Gym, Work, or Errands form the home page or notification.

02

Review & Edit

Tap any session to edit tag, notes, or times.

03

Get Suggestions

Get a notification to switch tags when in a certain location.

04

Dashboard

See time-per-tag totals for today and the week.

New Features 📈

Location-Based Suggestions

Get suggested tags via notifications based on proximity to a location set in the app.

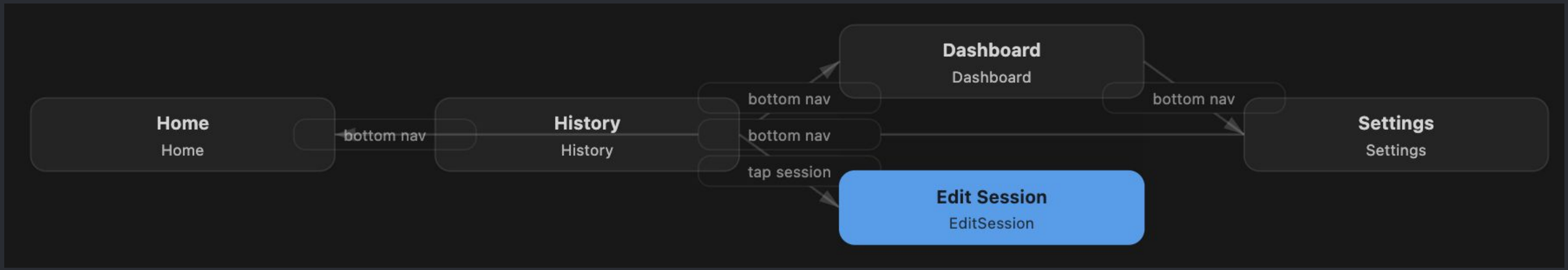
Persistent Notification

Change or log sessions directly from a notification. Select which tags are shown on the notification.

Custom Tags

Create custom tags and select which tags are shown on the home page.

Navigation


Dashboard
Dashboard
bottom nav
bottom nav
Home
Home
bottom nav
History
History
bottom nav
Settings
Settings
tap session
Edit Session
EditSession

MVVM Flow

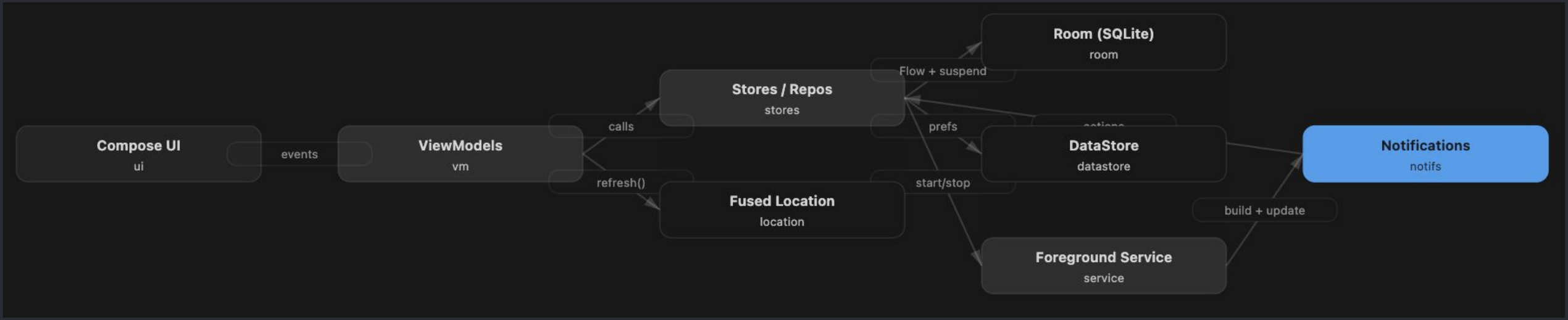

Compose UI
ui
events
ViewModels
vm
calls
refresh()
Stores / Repos
stores
Fused Location
location
Flow + suspend
Room (SQLite)
room
prefs
DataStore
datastore
start/stop
Foreground Service
service
build + update
Notifications
notifs

Table Schemas

Table: sessions

column	type	nullable	notes
id	Long	no	PrimaryKey (manual: time-based ids)
tag	String	no	Activity tag (includes Idle)
startTimeMillis	Long	no	Epoch millis
endTimeMillis	Long	yes	Null until completed
notes	String	yes	Optional text
edited	Boolean	no	Whether user edited entry

Table Schemas

Table: saved_locations

column	type	nullable	notes
id	Long	no	PrimaryKey autoGenerate
label	String	no	Human label ("Library")
latitude	Double	no	WGS84
longitude	Double	no	WGS84
suggestedTag	String	no	Tag proposed when inside radius
radiusMeters	Int	no	Default 150; constrained in UI

Debugging and Testing Strategy

- Used Compose Previews to tweak cards and dialogs without running the full app.
- Used in-memory Room tests to verify database relations and search queries.
- Tested edge cases (e.g., deleting a folder keeps entries consistent).
- Used Database Inspector and Logcat to debug data and UI issues.

LIVE DEMO

See It in Action 🗣️

A walkthrough of the core ClockIn experience, from starting a session to reviewing your day.

Future Features 📉

Google Calendar Sync

Connect your calendar and auto-suggest sessions based on scheduled tasks.

Lock Screen Widget

Persistent widget on the lock screen that allows you to change the current tag.

Descope: Automatic Locations

Gather frequent locations automatically.

Team Workflow

Sadid: data layer, Room database, notifications, location logic

Saksham: UI/UX, Compose screens, navigation, dashboard

Shared: architecture decisions, testing, debugging, final integration, presentation

AI Usage

- Used ChatGPT to help refine the proposal, narrow scope, explain Android concepts, debug issues, and generate some starter code for features like UI structure, state handling, and database logic
- AI was helpful for speeding up brainstorming, producing boilerplate faster, and giving quick explanations for new concepts
- Its limitations were that some suggestions were too generic or not fully correct, so we had to review, test, and fix errors

Thank You 🤩

ClockIn, because knowing where your time *actually* goes is the first step to owning it.